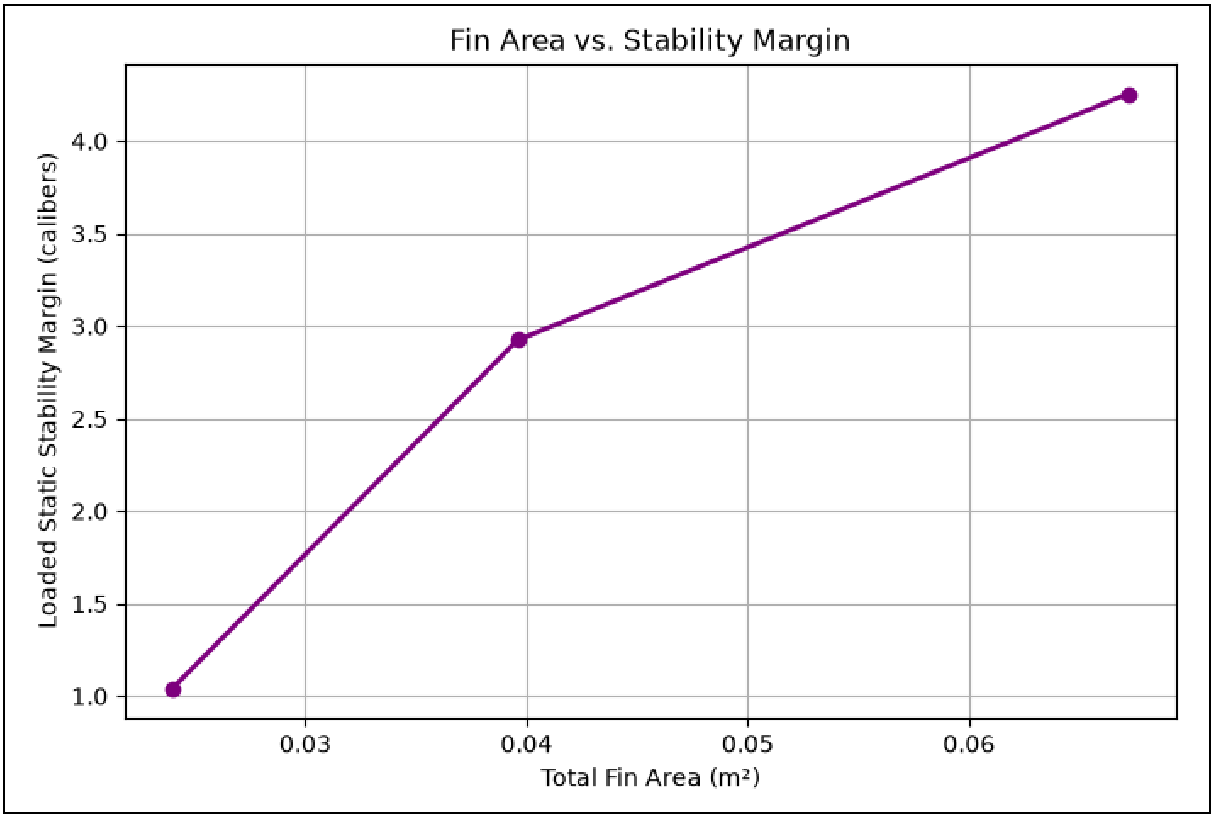
figure 1a



- the upward-sloping purple line illustrates a clear relationship: as the physical size (total area) of the rocket's fins increases, the rocket becomes significantly more aerodynamically stable
- moving from the smallest fin configuration on the left to the largest on the right drastically boosts the "Loaded Static Stability Margin" from roughly 1 caliber to over 4 calibers

output

Configuration	Fin Area (m^2)	Stability Margin (cal)	Apogee (m)
Case A (Small)	0.0240	1.04	3810.15
Case B (Medium)	0.0396	2.92	3808.91
Case C (Large)	0.0672	4.26	3808.21

- while larger fins (Case C) make the rocket much more stable, their extra drag and mass actually cause a slight drop in the rocket's maximum altitude (apogee) compared to the small fins (Case A)
- shows that more than doubling the fin area (from 0.0240 m^2 to 0.0672 m^2) costs the rocket nearly 2 meters of peak altitude (dropping from 3810.15 m to 3808.21 m)